LeWM reproduction

TwoRoom, PushT and Cube met our practical baseline check. Reacher fell short.

Environment	Paper	Primary	Repeats
TwoRoom	87%	90% (45/50)	74% / 90%
Reacher	86%	64% (32/50)	68% / 86%
PushT	96%	96% (48/50)	88% / 88%
OGBench Cube	74%	72% (36/50)	84% / 64%

Primary: seed 42. Repeats: seeds 43 and 44. Each score uses 50 cases. The predeclared check allowed a primary score up to 10 percentage points below the paper. This is a practical tolerance, not proof of equivalence.

What we ran. Frozen released LeWM checkpoints with the original Cross-Entropy Method (CEM) planner. Goals were 25 recorded actions ahead; the controller had 50 actions to reach them. All **15 panels passed the evidence audit**: 750 episodes and 37,500 controller actions, including the separately planned CEM sensitivity checks.

These are nearby-goal results from shared checkpoints. They do not establish safety, long-horizon performance or replication across training seeds.

What this means for our testing

- **Start from the tested baselines.** TwoRoom, PushT and Cube provide starting points for adapter studies. Reacher's shortfall needs diagnosis before gains there can be interpreted confidently.
- **Measure safe completion.** Compare adapters with native LeWM on matched cases and budgets, across multiple seeds. Report safe completion, violations and unfinished episodes together; test longer horizons separately.

Experiment completed 4 November 2026. Summary revised 5 November 2026.

Full results, uncertainty and runtime caveats | Videos by environment